Continual Learning and Affective State Identification Across World Languages

Anonymous ACL submission

Abstract

1 Introduction

ND:[Prior emotion detection work/why is emotion understanding important] Historically, research on sentiment and emotion detection has relied on small, predetermined label sets (e.g., Ekman (1984, 1992), Plutchik (1980)) that are applied universally across varying linguistic and cultural backgrounds. While recent work has considered larger emotion label sets (Wang et al., 2020; Mohammad and Kiritchenko, 2015; Lian et al., 2024), few studies have considered open-vocabulary settings grounded in natural language emotion expression.

ND:[Why affective states/open vocabulary] The predetermined label sets used in traditional emotion detection datasets have several limitations: (1) emotion taxonomies can struggle to capture nuanced emotions such as *grief* and(2) emotions are typically labeled by third-party annotators who bring their own cultural and (3) linguistic perspectives to the annotation task. To address these issues, we instead derive labels from natural language, avoiding annotator biases and allowing variation across speakers.

ND:[Summary of background on emotion variation] Perhaps most importantly, traditional emotion taxonomies may be applied universally, failing to give attention to variation in cognitive understanding of emotion terms, e.g., across cultural and linguistic boundaries (Jackson et al., 2019). Work has shown, for example, that *frustration* in English is closer to *anger* while in different European languages, it can denote other emotional experiences (Soriano and Ogarkova, 2025). Alternatively, work has shown that such linguistic and cultural variation does not extend to language models, finding that emotions like *pride* are primarily interpreted from a Western lens (Havaldar et al., 2023). Accordingly, it is vital to consider this variation when evaluating LLMs' understanding of emotions across languages.

In this work, we focus on evaluating multilingual LLMs' understanding of affective states–fine-grained and nuanced labels of emotions and feelings–across a variety of languages. We build on prior work (Deas et al., 2024) to collect a large-scale dataset of personal narratives with self-reported affective states, considering a broader set of linguistic patterns, sources, and languages. We then leverage this dataset toward two lines of work toward characterizing models understanding of emotions across languages: 1) studying training dynamics in this domain and 2) studying models' representations of affective state labels. We outline our contributions as follows:

1. We curate a **large-scale, native-speaker validated dataset for affective state identification, ND:[Need a name]**, covering **eight languages** with varying resourcedness and typologies.
2. We benchmark multilingual LLMs and study **cross-lingual training dynamics** on this task, finding that TODO: *[...Summarize trends]*
3. We use the collected benchmark to study **models' internal representations** of fine-grained affective states and **how those representations are impacted by training**. We find that TODO: *[Findings]*

2 Background and Related Work.

3 Data

ND:[(Nick) What is the goal of the data detection (i.e., language coverage, native speaker personal narratives, diverse linguistic contexts.]

ND:[Explicitly distinguish from A) other emotion datasets, B) MASIVE, and C) open-vocabulary annotations]

3.1 Languages

Language	Code	Family	Branch	Tier
English	eng	Indo-European	Germanic	5
Spanish	spa	Indo-European	Italic	5
French	fra	Indo-European	Italic	5
Romanian	ron	Indo-European	Italic	3
Persian	pes	Indo-European	Indo-Iranian	4
Hindi	hin	Indo-European	Indo-Iranian	4
Mandarin	cmn	Sino-Tibetan	Chinese	5
Indonesian	ind	Austronesian	Malayo-Polynesian	3

Table 1: Included languages with language family information, ISO 639-3 code and resource tiers (Joshi et al., 2020). Unless specified in the ISO code, the region or dialect is non-specific. ND:[Should add scripts here as well]

3.2 Data Sources

Source	Lang.	License
OSCAR	All	CC0
CC-100	All	CC0
Reddit	All	Public API
YouTube	All	Public API
HmBlogs	pes	Research
OpenSubtitles	hin	CC BY-SA
EmotionTalk	cmn	CC BY-NC-SA 4.0
NSMC[1]	kor	CC BY 2.0

Table 2: ND:[@Parmida: Table with sources across languages (like SemEval)]PS:[need to update ron/hin/kor/cmn data but placeholder for now]

ND:[@Parmida, Quick description/citation (if needed) of the broad sources we used (e.g., we can just group prior emotion datasets under one heading). Can point to an Appendix with specifics about preprocessing, special considerations, etc.] We collect formal and informal text with expressions of affective states from sources that fall into web-scale crawls, community platforms, and curated language-specific corpora; Table 2 lists the assignment of sources to languages. These source classes capture different styles of expression within each language, ranging from edited web prose to spontaneous speech.

3.3 Data Collection

Bootstrapping ND:[(Nick) Briefly describe/cite MASIVE]

Prior work (Deas et al., 2024) introduced a data collection approach for gathering open-ended affective states instead of fixed emotion classes. We extended the methodology by implementing

Lang	Context	Label
eng	...I feel __ in myself that I did not report him to the police or have a pepper spray...	dissapointed
spa	...Que no es como si estuviera __ por conseguir novia pero preferiría amigos...	desesperado *desperate*
pes	احساس__و ناامیدی داشتم...	غم *grief*
hin	bahut __ hoon aaj yaar...	sad
ron	Sunt __ să-mi vând toate lucrurile pt bani de plecare...	disperat *desperate*

Table 3: Abbreviated context and label examples. PS:[I will update this later to add more languages and the somatic-idom examples with a new column for pattern. I think we should also add the translation of the non english one as well]

Language	# Train	# Val	# Test	# Aff. States	Avg. Len.
English	∼93	∼10k		1,627	∼311
Spanish	∼30			1,002	∼241
Persian	∼				
Hindi	∼				
Romanian	∼				
French	∼				
Korean	∼				
Mandarin	∼				

Table 4: Summary data statistics for each language including total size (*# Docs*), number of unique labels (*# Aff. States*), and text lengths are measured in mT5 tokens (*Avg. Len.*).

a shared bootstrapping pipeline: (1) per language seed affective states from emotion lexica or translated emotion categories, (2) extract first-person contexts via conjunction patterns (e.g., "I feel X and Y" where X is the seed) from corpora, (3) add newly discovered candidates to seed list and iterate, for a fixed number of rounds, and (4) validate candidates through LLM and human annotation. For languages that express emotion through somatic idioms, we extended the pipeline by extracting context view somatic patterns: e.g., Persian (del-am X shod), where the body part *del* (heart") in relation to X is used to express an emotion, and the idiom as a whole expresses the affective state (Sharifian, 2008; Rezai and Samenian, 2023). In addition to the 8 listed languages, we capture transliterated and code-switched text, e.g., Hindi-English (*Hinglish*). After extraction, every affective state is masked(both seed and discovered conjuncts; somatic idioms masked as the whole phrase), examples are deduplicated, and split into train/val/test as seen in Table 4.

3.4 Language-Specific Methods

TODO: *[@All: Temporary section for now so we have all of the details for each language that we can merge/organize between main paper and ap-*

pendix. Please include details like those that are skeletoned out for English.]

English. TODO: *[Where did you get your initial seed set from? (i.e., did you start with Ekman or another set?)]*

TODO: *[What templates did you use?]*

- Template 1 (*rough translation*)
- Template 2 (*rough translation*)
- ...

TODO: *[Any other necessary changes for your data? (i.e., did bootstrapping not work? Anything else you did differently/additionally? Why?]*

Spanish.

French. Initial seed from reddit esp. r/besoindeparler; template is je me sens X et Y, Y is new candidate state. ”heureux”, ”heureuse”, ”triste”, ”furieux”, ”furieuse”, ”effrayé”, ”effrayée”, ”dégoûté”, ”dégoûtée”, ”surpris”, ”surprise”

Romanian. We first construct a Romanian affective state vocabulary. We map the WordNet-Affect synsets (Strapparava and Valitutti, 2004) from WordNet 1.6 to WordNet 3.0 and match them against the English glosses in RoEmoLex (Lupea and Briciu, 2019). WordNet-Affect contains 798 synsets, of which 788 can be mapped to WordNet 3.0. The cross-reference with RoEmoLex produces 479 matched entries. We manually inspect these matches, remove verbs and multi-word expressions, and correct participial forms. This gives an initial list of 237 Romanian terms: 92 adjectives, 131 nouns, and 14 adverbs. We then add 148 manually collected terms that were missed by the automatic mapping. The resulting seed contains 385 terms.

We next use the collected Romanian corpora to extend this list. We first apply conjunction templates based on the method used by MASIVE (Deas et al., 2024). These templates search for constructions such as *mă simt X și Y*, where one of the two terms is already in the seed. We use only unambiguous forms of *a simți* and exclude constructions based on *a fi*, since forms such as *sunt* can refer either to the first-person singular or the third-person plural. The conjunction method does not produce any reliable additions after manual inspection.

We also search for terms that occur in five Romanian distributional templates: *un sentiment de X*, *sentimentul de X*, *emoție de X*, *o senzație de X*, and *copleșit/copleșită de X*. The candidates are manually inspected. This step adds 139 terms, most of which are nouns, and increases the final seed from 385 to 524 terms.

For the final extraction, we use 20 Romanian patterns. Ten patterns are based on forms of *a simți*, while the remaining patterns cover constructions based on *a fi*, *a se face*, *a-i fi*, and *a avea*. We expand adjective lemmas to masculine and feminine singular forms and allow matches with and without Romanian diacritics. The complete list of patterns is given in Appendix C.

The extraction gives 5,227 candidates from the six smaller datasets, 29,452 from Filmot, and 100,000 from FULG. We remove duplicate sentences using the priority order smaller datasets, Filmot, and FULG. This leaves 129,700 candidates: 4,905 from the smaller datasets, 28,314 from Filmot, and 96,481 from FULG.

Persian.

- **Initial Seed Construction.** 12 initial seeds: the six Ekman categories (Ekman, 1992) in both noun form (شادی, غم, خشم, ترس, تعجب, نفرت) and adjective form(خوشحال, غمگین, خشمگین, ترسان, متعجب, متنفر). Including both forms is required because of how Persian handles emotional predication. Specifically, Persian relies heavily on light-verb complex predicates; the frame ”I feel”(«احساس X می‌کنم») requires a nominal argument (Karimi-Doostan, 2005; Folli et al., 2005). While, adjectival states must predicate through a first-person copula or inchoative verb like «X شدم» (Mostafavi, 2015).
- **Template used for data extraction.**
  - **Noun frame.** «احساس X و Y می‌کنم» (*lit. “I make the feeling of X and Y” ≈ “I feel X and Y”*)
  - **Adjective frame.** « X و Y شدم » (*“I became X and Y”*)
  - **Somatic-idiom frame.** «دلم X شد» / «دلم گرفت» (*“my heart became X” / “my heart seized” ≈ “I felt down”*). Body part + 1sg possessive enclitic ـم + verb. Here, the enclitic satisfies the required first-person constraint. Because these expressions are non-compositional, the idiom is masked as a single, contiguous phrase (Sharifian, 2008; Rezai and Samenian, 2023).

Hindi.

Mandarin.

- **Initial seed construction.** Started with Chinese translations of the six Ekman basic emotion categories (Ekman, 1992). Initial seed set included common Mandarin forms such as 开心/高兴 (happy), 难过/伤心 (sad), 生气/愤怒 (angry), 害怕/恐惧 (afraid), 惊讶 (surprised), and 恶心 (disgusted).

- **Templates used for data extraction.** We used a variety of Mandarin templates such as self-state and conjunctive/contrastive templates. These were used in different stages of the affective state extraction pipeline. Example templates include:

  - Self-state: 我很 X (*I am very X*)
  - Self-state: 我有点 X (*I am a little X*)
  - Self-state: 我觉得 X / 我感觉 X (*I feel X*)
  - Conjunctive: 我又 X 又 Y (*I am both X and Y*)
  - Contrastive: 我很 X 但也很 Y (*I am very X but also very Y*)
  - 这让我很 X (*This makes/made me very X*)

- **Other language-specific changes.** Bootstrapping worked for Mandarin, but several additional cleanup steps were needed. First, we normalized text toward Simplified Chinese to reduce duplication between Traditional and Simplified variants. Second, for YouTube/Filmot transcripts, we restored punctuation before segmentation because many subtitle files were under-punctuated. Third, we added an extra label-level audit before dataset creation because some Chinese strings that look affective in isolation are often non-affective within specific contexts. For example, 希望 can mean "hope" or "want," 感谢 is often a thanking formula, 安全 often refers to physical/security safety, and 对不起 is often an apology formula. We sampled matched examples for high-risk labels and removed labels judged too noisy before creating masked examples.

Indonesian.

3.5 LLM Filtering

Romanian. We score the 129,700 Romanian candidates with `Qwen/Qwen3.5-9B`.[2] The prompt is written in Romanian and defines four scores. A score of 0 means that the matched term does not describe the speaker's affective state. A score of 1 indicates an uncertain or weak use, a score of 2 indicates a plausible affective state, and a score of 3 indicates a clear affective state of the speaker. The prompt also contains seven Romanian examples.

We mark the extracted term with a `span` element and provide at most 5,000 characters of context around it. Decoding is deterministic, with a temperature of 0. The model returns 9,552 examples with score 0, 16,348 with score 1, 30,373 with score 2, and 73,427 with score 3. No responses fail to parse.

3.6 Data Validation

Romanian. We sample 200 Romanian candidates for human validation, with 50 examples from each LLM score. The examples are sampled proportionally from the three source groups. Two native Romanian speakers annotate the examples using the same 0–3 scale and the same seven examples as the LLM prompt. A subset of 105 examples is annotated by both speakers.

The weighted Cohen's kappa between the two speakers is 0.65. The Spearman correlation between the mean human score and the LLM score is 0.70. Human annotators confirm the matched term as an affective state in 4.0% of the score-0 examples, 29.6% of the score-1 examples, 46.7% of the score-2 examples, and 91.3% of the score-3 examples. We therefore retain only examples assigned a score of 3.

This filtering step leaves 73,427 examples. We then re-run the original Romanian extraction pattern on each example and verify the exact position of the matched term before replacing it with `[MASK]`. We verify 70,289 examples, corresponding to 95.7% of the filtered data. After normalization, these examples contain 372 distinct affective state terms. We use this verified version for the experiments.

[2] https://huggingface.co/Qwen/Qwen3.5-9B

ND:[Human annotations]

4 Experimental Methods

4.1 Models

ND:[Qwen3.5, gpt-oss?]

4.2 Metrics

Top-k Accuracy.

Top-k Similarity.

4.3 Zero-Shot Evaluation

ND:[...]

Figure 1: ND:[Prompt used for evaluation]

4.4 Training Dynamics

ND:[@Both: For now, quick details on how the task was structured for instruct-tuned models/how you did finetuning for the group presentation experiments. Don't need to include specific hyperparameter values like max length or lora params (if used) unless they are relevant to interpretting the results, just higher-level mentions like "Constrained generation length" and reference Appendix D with the specifics.]

4.5 Activation Space Structure

5 Results

5.1 Zero-Shot Evaluation

5.2 Finetuning Evaluation

ND:[What all do we want to show in the main paper here?]

5.3

6 Conclusion

Limitations

Ethics Statement

References

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Language	Model Checkpoint	Citation	Why?
English	ND:[copy huggingface checkpoint]	ND:[copy citation]	ND:[Any reason this model is better than others? (e.g., benchmark performance, etc.)]
Spanish			
French			
Romanian	OpenLLM-Ro/RoLlama3.1-8b-Instruct	(?)	For Romanian, we use RoLlama3.1-8B-Instruct (?). The model is an instruction-tuned Romanian adaptation of Llama 3.1 8B and falls within the 7–9B parameter range used in our comparison.
Persian	universitytehran/PersianMind-v1.0	(Rostami et al., 2024)	**PersianMind (7B):** Adaptation of Llama-2-7B. Has paper publication(arxiv only), vocab expanded by 10k tokens. **Caveat:** Older base model and TARAZ found to not be as good as other models.
	MehdiHosseiniMoghadam/AVA-Llama-3-V2	(Iranmanesh et al., 2026)	**AVA-Llama-3 (8B):** Adaptation of Llama-3-8B. SOTA Llama 3 backbone; formally evaluated on the TARAZ cultural short-answer benchmark and found to be the best open Persian finetuned one. **Caveat:** No standalone model paper.
Hindi			
Mandarin	Qwen/Qwen2.5-7B-Instruct	(Yang et al., 2024)	**Qwen2.5-7B-Instruct:** Chinese-native Qwen-family instruction-tuned model. Improved pre-training and post-training over earlier Qwen models, including scaling pretraining data to 18T tokens.
Indonesian			

Table 5: LLMs created specifically for each language. ND:[@All: Need to be 1) instruction-tuned, 2) have a 7-9B variant. Preferably a model that is relatively recent (last 3-5 years), has an attached publication in a peer-reviewed venue (i.e., not just arxiv), and is an adaptation of a Llama, Qwen, or Mistral model.]

Language	Model Name	Checkpoint	Citation	Why?
English				
Spanish				
French				
Romanian	`dumitrescustefan/bert-base-romanian-cased-v1`	BERT-base	(Dumitrescu et al., 2020)	For the Romanian encoder, we use Romanian BERT (Dumitrescu et al., 2020). It is a cased, Romanian-only BERT-base model with 12 layers and a hidden size of 768. The cased model also preserves Romanian diacritics.
Persian	AriaBERT	`ViraIntelligentDataMining/AriaBERT`	(Ghafouri et al., 2023)	Upgraded RoBERTa-base model using a BPE tokenizer. Outperforms ParsBERT on downstream tasks by up to 3%.
Hindi				
Mandarin	BGE-large-zh-v1.5	`BAAI/bge-large-zh-v1.5`	(Xiao et al., 2023)	BERT/feature-extraction based, 1024-dimensional, and evaluated on C-MTEB, a Chinese text embedding benchmark.
Indonesian				

Table 6: Embedding models created specifically for each language. ND:[@All: should be encoder-only (i.e., bert/roberta-based)]

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A Model Details

B Data Source Details

Sources

Licenses

C Query Templates

Romanian extraction patterns. The Romanian extractor uses the following pattern families:

1. Present and past forms: *mă simt X*, *mă simțeam X*, and *m-am simțit X*.
2. Future and conditional forms: *mă voi simți X*, *o să mă simt X*, and *m-aș simți X*.
3. Subordinate and nominal constructions: *să mă simt X*, *simt că X*, *simt [NOUN]*, and *simțeam [NOUN]*.
4. Copular constructions: *sunt X*, *eram X*, *am fost X*, and *o să fiu X*.
5. Other affective constructions: *mă fac X*, *îmi este X*, *îmi era X*, *mi-e X*, *am [NOUN]*, and *aveam [NOUN]*.

For adjective terms, the extractor generates masculine and feminine singular forms. Diacritic and non-diacritic forms are treated as variants of the same term. Because *sunt* can mean either “I am” or “they are”, examples extracted with this pattern are retained only after the LLM validation step.

D Generation and Finetuning Details

ND:[We can include the specific values of hyperparameters and such here rather than main paper.]

	fa	**ro**	**hi**	**fr**
Seeds (initial $\rightarrow$ final)	12 $\rightarrow$ TODO	385 $\rightarrow$ 524	64 $\rightarrow$ 133	11 $\rightarrow$ 298
Frames	3 families	20 patterns	3 stages	2 templates
Conjunction bootstrapping	(3 iter)	(failed)		(5 iter)
Somatic channel	(template)		(detection)	
Masked-LM expansion	(validation)		(MuRIL)	
Distributional patterns		(5 templates)		
Sources	3	8	7	3
LLM filter (0–3, open)	TODO	(Qwen3.5-9B)	partial	partial
Human validation	TODO	(κ_w=.65)	planned	
Examples / labels	TODO	70,289 / 372	22,432 / 120	76,927 / 298

Table 7: Per-language breakdown TODO caption text.